Heritage Lens Agent — Actionable Project Document

For NotebookLM + Presentation Preparation Created: 30 March 2026

THE PITCH (one sentence)

> "Most AI systems optimise for answers. This one makes the construction of those answers visible and accountable."

WHAT WE ARE BUILDING

A domain-aware research agent for specialised archives where provenance, gaps, and contested interpretations matter. It takes a research question, retrieves relevant sources, synthesises a grounded answer, evaluates its own output, and — critically — flags what it cannot confirm.

The key difference from standard RAG systems: Standard systems hide uncertainty. Heritage Lens Agent surfaces it as structured information.

THE THREE-LAYER OUTPUT (non-negotiable)

Every response contains exactly three layers. This is the product.

Layer 1 — The Grounded Answer Direct response using only retrieved sources. Any general knowledge is explicitly labelled `[BACKGROUND — not retrieved]`. If evidence is weak, this is stated plainly.

Layer 2 — Source Grounding Full attribution: source name, source type, institutional origin. If nothing valid was retrieved: *"No reliable sources retrieved."*

Layer 3 — Epistemic Transparency The differentiator. Structured, scannable, tied to the actual retrieved data:

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Source bias:
- Dominant source types: [e.g. 4/5 Western academic papers]
- Institutional patterns: [e.g. UNAM, Italian university press]

Absences:
- Missing knowledge types: [e.g. no oral traditions, no community archives]

Interpretive limits:
- [specific to this query, not generic]

Confidence: HIGH / MEDIUM / LOW
- Justified by strength and diversity of retrieved sources
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Hard rule: Layer 3 must be tied to the actual retrieved chunks. Generic disclaimers are a failure state, not a feature.

THE CORPUS (our actual data)

What we have:

- Ella's degree thesis (primary source)
- Professor's published book (secondary source)
- Both covering Mesoamerican cultural heritage / Ruta de la Obsidiana / Olmec context

Why this matters: This is not placeholder data. This is intellectually owned, metadata-rich material. We know exactly:

- Source type (academic thesis, peer-reviewed book)
- Institutional origin (Italian university)
- Language of origin
- Cultural perspective (Western academic archaeology)
- What is present — and what is absent (indigenous oral traditions, community knowledge)

This makes Layer 3 honest rather than simulated. When the system says *"this corpus reflects Western academic archaeology"* — that is a true, verifiable statement, not a generic disclaimer.

Demo query (confirmed):

> *"What was the ritual function of obsidian at Olmec ceremonial sites?"*

Contingency query (for weak retrieval demo): Choose a topic where the thesis/book coverage is thin. Show the system expanding the Absences section rather than confabulating an answer. This turns weak retrieval into a feature demonstration.

THE TECHNICAL PIPELINE

Minimal viable flow (what must work before anything else):

1. Load documents → chunk them → attach metadata
2. User asks a question
3. System retrieves top 3–5 chunks
4. LLM generates 3-layer response
5. Judge (second LLM call) evaluates whether Layer 3 is specific or generic
6. Output displayed in 3-panel Streamlit UI

Stack:

- LLM: GPT-4o (OpenAI API)
- RAG: LangChain or LlamaIndex (buzman's call)
- Vector DB: TBD — Chroma / Pinecone / Weaviate
- Agent orchestration: openclaw architecture (buzman's existing system)
- UI: Streamlit, 3 visible panels
- Judge layer: second GPT-4o call

Metadata schema (each chunk must carry):

- `source_name`
- `source_type` (thesis / book / excavation report / etc.)
- `institution`
- `date`
- `language_of_origin`
- `cultural_perspective`

Why metadata matters: Layer 3 cannot function on raw text chunks. The agent needs to evaluate the *shape* of its sources, not just their content.

THE JUDGE LAYER

A second LLM call that runs after the primary response. It checks Layer 3 only.

Evaluates:

1. **Specificity** — are claims tied to retrieved data or generic?
2. **Relevance** — does the analysis reflect this specific query?

3. **Honesty** — does it acknowledge weak or missing data?

Output: `VALID` or `WEAK` + two-sentence explanation.

If `WEAK` → the system regenerates Layer 3 before showing the user.

This is what separates the project from a prompt engineering exercise.

FAILURE AS A FEATURE

When retrieval is insufficient:

- Layer 1 states this explicitly: *"Retrieval is insufficient to answer definitively."*
- Layer 3 expands the Absences section
- System does NOT confabulate

This is not a bug. This is the most honest thing the system can do — and for Mission 4 judges, it is the most powerful moment in the demo.

> *"The absence of evidence becomes the evidence."*

THE TEAM

Person	Role	What they own
Ella	Project Lead — concept, epistemic design, pitch	System logic, Layer 3 enforcement, demo narrative
buzman	Lead Engineer — agent pipeline, LLM integration	RAG pipeline, metadata schema, judge loop, orchestration
J Lee	UI & Demo Lead — Streamlit, demo flow	3-panel interface, demo clarity, project spec
Nikhilesh	QA & Documentation — query testing, submission docs	Breaking the system, documenting outputs, submission

Available hours:

- buzman: fully available from April 5th (London)
- Nikhilesh: 11:30am–6:30pm London time (India)
- J Lee: busy until April 4pm, then available
- Ella: driving the project now, coordinating the build phase from April 5th

Coordination:

- WhatsApp: daily quick updates
- Google Drive: shared project folder, documents, decisions log
- GitHub: all code

BUILD PRIORITIES (in order)

Before April 5th (Ella):

1. Set up GitHub repo with folder structure
2. Set up Google Drive shared folder
3. Prepare corpus — thesis + book ready to chunk
4. Confirm demo query and contingency query
5. Brief J Lee on Streamlit layout (3 panels, nothing cluttered)

April 5th–8th (full team):

1. buzman: minimal pipeline running end-to-end (load → chunk → metadata → retrieve → LLM → output)
2. J Lee: Streamlit UI with 3 panels displaying the output
3. Ella: refine Layer 3 prompt until outputs are specific, not generic
4. Nikhilesh: test queries, document what breaks, report to team

April 8th–10th (polish + submission):

1. Run demo query end-to-end, smooth the flow
2. Add contingency/failure demo
3. Record 5-minute demo video
4. Finalise submission document

Non-negotiable rule: One working query with visible 3-layer output by April 8th. Everything else is secondary.

THE DEMO (60–90 seconds)

Opening:

> *"Most AI systems give answers. This one shows you how those answers are constructed — and what they leave out."*

Flow:

1. Type the demo query live
2. Layer 1: *"A grounded answer based only on retrieved sources."*
3. Layer 2: *"Every claim is traceable."*
4. Layer 3 (slow down, let the screen do the work): *"This is the key difference. It tells you what kind of knowledge dominates the answer — and what's missing from the dataset."*
   > *"For example — no indigenous oral accounts are present here. So this isn't a complete picture. It's a documented one."*
5. Close: *"We're not building smarter answers. We're building more accountable ones. Because in cultural heritage, what's missing matters as much as what's present."*

Delivery rules:

- Never mention RAG, embeddings, or vector databases to non-technical judges
- Pause at Layer 3
- Frame as *"look what this reveals"* not *"look what I built"*

POSITIONING

What to say:

- *"Accountable AI for specialised research"*
- *"We encode anthropological thinking into system behaviour"*
- *"An agent that doesn't just answer — it checks itself"*

What not to say:

- *"Ethical AI"* — too generic
- *"Digital colonialism"* — use once, deliberately, not as a refrain
- Any technical jargon to non-technical judges

Our edge: Most Mission 4 entries will be abstract policy arguments. We have a working system, real data, and a genuine intellectual position about whose knowledge frameworks govern AI systems. That combination is rare.

KEY PHRASE FOR JUDGES

> *"Instead of pretending to know everything, the system shows you the limits of what it knows."*

Document prepared by Ella + Claude (Anthropic), 30 March 2026 This is a living document — update as decisions are made